

SLURRY_FREE_ACID Modeling and Process Intelligence

Project Progress and Model Evaluation

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OCP Group — Internship Progress Update

August 27, 2026

Current Project Status

- I built and evaluated three distinct modeling setups for SLURRY_FREE_ACID using one month of process data.
- I compared each setup against simple baselines using chronological (time-respecting) validation.
- I selected working candidates for two of the three use cases and rejected one additional upgrade after testing it.
- I built an interactive dashboard that runs the saved models and shows prediction reliability information.
- **Status:** internally validated on retrospective data. Not yet operationally validated on new, independent data.

Dataset and Evaluation Design

Dataset

- 44,640 chronological observations
- ≈ 31 days, January 2026
- One observation per minute
- 21 original process measurements
- Target: SLURRY_FREE_ACID

Chronological split

- $\approx 70\%$ train / 15% validation / 15% test
- No random shuffling
- All preprocessing (imputation, scaling, lag selection, filtering, clipping) fitted on training data only

Why chronological, not random

The 44,640 rows are **not** 44,640 independent experiments — they are one continuous month with strongly dependent adjacent minutes.

Feature preparation (brief)

I built four feature sets as the experimental foundation: A — Raw (21 features), B — Temporal (162 retained), C — Process-aware (58 retained), D — Full (178 retained). I won't detail the engineering here — it's covered in the notebooks.

Three Operational Prediction Cases

This distinction matters for how any of these models could actually be used.

Case	Required input	Predicts	Purpose	Main limitation
1. Direct forecast	Process measurements up to t	$y(t + h)$	Early warning, $h=1/5/10/15$ min	Must beat persistence to be useful
2. Delta forecast	Process measurements + current/past y	$\Delta y(t, h)$, reconstructed as $y(t) + \Delta y$	Short-horizon early warning	Needs $y(t)$ at inference
3. Virtual sensor	Process measurements only	$y(t)$ (current, estimated)	Soft sensor when no lab/analyzer value exists	Estimates <i>now</i> , not the future

Direct Forecasting and the Persistence Result

Notebook 03 — all 16 combinations (4 horizons $\times$ 4 feature sets)

- Baselines: training-target mean, and persistence ($\hat{y}(t+h) = y(t)$) when $y(t)$ was available.
- Models tested: Ridge, Extra Trees, Histogram Gradient Boosting.

Result

The direct future-level models learned the general operating level of SLURRY_FREE_ACID, but did **not consistently outperform persistence**.

Why

SLURRY_FREE_ACID changes relatively slowly over much of the month, so the current value $y(t)$ is already a very strong predictor at short horizons.

Consequence: this motivated predicting the *change* instead of the full future level — the approach in Notebook 04.

Target-Anchored Delta Forecasting (Notebook 04)

Reformulation

$$\Delta y(t, h) = y(t + h) - y(t)$$

- Model learns only the movement.
- Best candidate: **Ridge**, hybrid/full feature configuration (process measurements + target history).

Example

current $y(t) = 11.20$

future $y(t + h) = 11.35$

model only needs to learn: $+0.15$

Result

This model beat persistence at **all four horizons**, with an improvement of roughly **19.8%–34.9%** across horizons.

Limitation

Requires $y(t)$ at inference — cannot be used on a period with no SLURRY_FREE_ACID measurement available at all.

Target-Free Virtual Sensor (Notebook 05)

Contract

- Training: X = process features, y = current SLURRY_FREE_ACID (target excluded from inputs).
- Inference: a new CSV with process measurements only can receive current free-acid estimates.
- Models evaluated: Ridge, PLS, KNN, Extra Trees, HGB, XGBoost, LightGBM, and weighted blends.
- Best: **50/50 Ridge + LightGBM blend** — Ridge for stable linear structure, LightGBM for nonlinear thresholds/interactions.

Retrospective test performance

MAE	0.2490
RMSE	0.3014
R^2	0.5583

$R^2 \approx 0.558$ means the model explained about 55.8% of target variance relative to a constant-mean reference — *not* “55.8% accuracy.”

Expanding Validation and the Notebook 06 Upgrade

Four expanding chronological validation blocks, each $\approx 1,657$ rows, training set growing each time (30,940 $\rightarrow$ 35,911 rows).

Candidates tested: stronger/global Ridge, recency-weighted Ridge (half-life 3/7/14 days), clustered Ridge experts (3 or 5 KMeans regions), LightGBM variants, Ridge–LightGBM blends.

Candidate	Mean RMSE	RMSE SD	Worst-block RMSE
70% Ridge / 30% LightGBM	0.3191	0.0762	0.4236
50% Ridge / 50% LightGBM	0.3201	0.0686	0.4124
5-cluster Ridge (selected)	0.3280	0.0396	0.3522

Selection and result

I selected **5-cluster Ridge** as the stability-first candidate: it has the lowest validation RMSE SD (0.0396) and best worst-block RMSE (0.3522). This accepts weaker mean RMSE than the 70/30 blend. After refitting on train+validation, retrospective test: MAE 0.2354, RMSE 0.2977 ($\approx 1.23\%$ below Notebook 05), R^2 0.5692. The clusters are statistical operating regions, not confirmed physical regimes.

Regime-Aware Hybrid: Accuracy vs. Stability (Notebook 07)

Notebook 06 selected 5-cluster Ridge for stability. Notebook 07 combines it with the best-mean **70/30 Ridge–LightGBM** accuracy anchor:

$$\hat{y} = (1 - w) \cdot \hat{y}_{\text{global}} + w \cdot \hat{y}_{\text{local}}$$

Configuration	Mean RMSE	RMSE SD	Worst-block RMSE
Global only ($w=0$)	0.3191	0.0762	0.4236
80% local (selected)	0.3213	0.0413	0.3548
Local only ($w=1$)	0.3280	0.0396	0.3522

Selected: 80% local hybrid, vs. global

Against the 70/30 accuracy anchor, the selected hybrid accepts $\approx 0.69\%$ higher mean RMSE while reducing worst-block RMSE by $\approx 16.25\%$ and RMSE SD by $\approx 45.78\%$. Retrospective test: MAE 0.2324, RMSE 0.2903, R^2 0.5902; RMSE is $\approx 2.47\%$ below Notebook 06's clustered model. *This remains retrospective evidence.*

JITL Experiment — and Why It Was Rejected (Notebook 08)

Idea: instead of 5 fixed clusters, find query-specific historical neighborhoods and fit a small local Ridge model per query, blended with the global model.

- Similarity spaces: PCA (10/20 comp.), PLS (5/10 comp.)
- Neighbor counts: 100 / 250 / 500; weighting: uniform / inverse distance
- Local contribution: 25% / 50% / 75% / 100%
- 97 candidate configurations, evaluated across the same 4 validation blocks

	Mean RMSE	RMSE SD	Worst-block RMSE
Best JITL (PLS-10, 500 nb., uniform, 25% local)	0.3254	0.0799	0.4353
70/30 global accuracy anchor	0.3191	0.0762	0.4236

Decision: rejected

Best JITL was $\approx 1.98\%$ worse on mean RMSE and $\approx 2.75\%$ worse on worst-block RMSE than the 70/30 global anchor, with higher variability and cost. Validation therefore retained the global anchor. Likely limits: similarity in measured variables may not imply the same free-acid relationship, and one month provides limited operating diversity.

Final Modeling Decision

Candidate	Role	Retrospective metrics	Status
Ridge delta model	Target-anchored forecasting	Beat persistence, all horizons, $\approx 19.8\%$ – 34.9% better	Selected
5-cluster Ridge	Stability-first target-free	MAE 0.2354, RMSE 0.2977, R^2 0.5692	Selected
80% local regime hybrid	Balanced target-free alternative	MAE 0.2324, RMSE 0.2903, R^2 0.5902	Selected (alt.)
JITL	Query-specific local model	Worse mean & worst-block RMSE, higher variability	Rejected
Operational model	Deployed / control-grade	—	Not selected — needs external validation on a later labeled month

Dashboard and Row-Level Reliability Outputs

Capabilities

- Default January dataset or CSV upload
- Process-variable exploration, correlations, relationship explorer
- Candidate transition detection (top 5% multivariate change score — analytical label, not confirmed plant events)
- Target-free current-value prediction & target-anchored forecasts (1/5/10/15 min)
- Downloadable predictions, drift/familiarity info, report generation
- Loads saved artifacts only — **does not retrain**; first 30 uploaded rows are warm-up (no prediction)

Row-level confidence, on hover/focus

Predicted free acid: 11.42

90% interval: 10.98–11.86

Process familiarity: low

Confidence status: verification recommended

Caution

The 90% interval is an empirical, within-month interval. Familiarity compares inputs to the training envelope — it is *not* a probability of correctness, and low familiarity does not prove the prediction is wrong.

Limitations, Open Questions, and Next Steps

Main limitations

- Only one month of labeled data; adjacent minutes are dependent
- No independent seasonal/product-grade validation
- January test period has already been inspected
- Uncertain engineering units for several variables (acid flows, ground-phosphate, fuel, steam, wash-liquid, recycling)
- Process-aware ratios remain provisional; KMeans clusters are not confirmed physical regimes
- No causal interpretation from model importance; no control recommendation without confirmed controllable variables
- Empirical intervals not yet externally calibrated
- Target-free accuracy cannot be measured on a new CSV without some real reference values

Next steps

- 1 Obtain a later labeled month (process vars + measured free acid, ideally + grade/event info)
- 2 Freeze current candidates before inspecting it
- 3 Evaluate without retuning: MAE, RMSE, R^2 , bias, transition/extreme-value errors, missing-sensor behavior, drift, interval coverage
- 4 Compare the three selected candidates on that data
- 5 Calibrate uncertainty intervals on genuinely later data
- 6 Get process engineers to validate units, proxies, controllable variables, regimes, event windows
- 7 Only then select an operational prototype

Conclusion

- Predicting the future *change* beats persistence when the current target is available.
- Process-only current-value estimation (virtual sensing) is feasible within this month's data.
- Five-cluster Ridge is the stability-first target-free choice; the 80% local hybrid gives the strongest retrospective balance of accuracy and stability.
- Just-in-time learning did not justify its added complexity and was rejected.
- The dashboard provides process exploration, two prediction modes, row-level reliability context, and analysis-ready exports.
- **The project is ready for external shadow testing, not operational control.**
- Next requirement: a genuinely later labeled month, plus process-engineer validation.